SOFTWARE ENGINEERING - PROJECT PROPOSAL

NewEra POS

A point-of-sale system for a retail store - sales, returns, rentals and payment processing



Software Engineering · Beihang University (BUAA)

September 2026

Alfonso Pinto 阿方索



MAJOR

Double Degree in Computer Science and Business

ABOUT ME

I am from Spain and I am one of the few Undergrads in the class, and still, I am one of the few with a Software Engineering background.

STRENGTHS FOR THE PROJECT

- C++, Java and Python
- Knowledge in Software Engineering
- Extroverted and ease to talk

Haifa Chkoundali 海法



MAJOR

Telecommunications / Computer Science

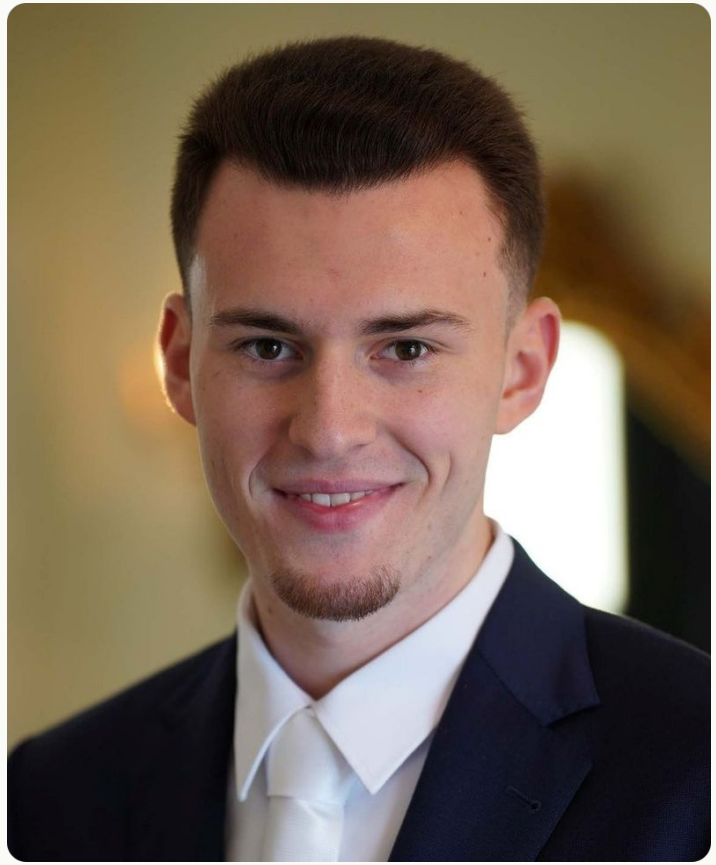
ABOUT ME

Originally from Tunisia and currently studying engineering at INSA Lyon in France, I am particularly interested in cybersecurity and its relationship with telecommunications.

STRENGTHS FOR THE PROJECT

- Full-stack dev
- Software engineering & coding practices
- AI / LLM integration (Conversational agents)
- Creative problem-solving & innovation

Borja Nicolás 博尔加



MAJOR

Computer Engineering — Major in Cyberspace Security

ABOUT ME

I am passionate about cybersecurity and how to keep systems and data safe. My background is in computer science studies, which gives me a solid foundation in programming, software design and how systems work.

STRENGTHS FOR THE PROJECT

- Experience in Java programming
- UML modeling
- Teamwork
- Documentation and presentation

Yasmine Chkoundali 雅斯敏



MAJOR

Telecommunications & Digital Services Engineering

ABOUT ME

I'm from Tunisia, a small but beautiful country in North Africa, and I'm currently studying engineering in France at INSA Lyon. I'm interested in how technology can make everyday life easier

STRENGTHS FOR THE PROJECT

- UI/UX experience — designing interfaces that are clear and intuitive for the end user
- System design / overview — able to see and design the whole information system (SI), not just isolated pieces
- Programming / coding

Josué Rodríguez 江钧刚



MAJOR

Cyberspace Security

ABOUT ME

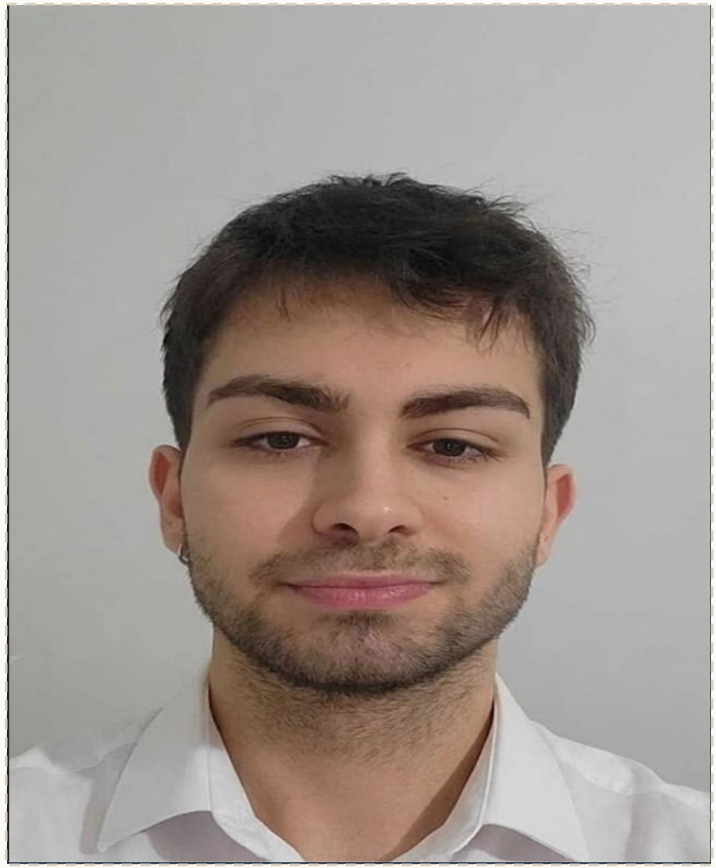
I’m from Ecuador. I’m passionate about privacy, data protection, compliance, and incident response. My goal is to connect legal and technical perspectives, bringing ethics and risk awareness to the table.

STRENGTHS FOR THE PROJECT

- Secure design, privacy & compliance focus
- Requirements traceability & documentation
- Incident response & risk analysis
- Translating legal requirements into technical language

THE TEAM · MEMBER 6 OF 6

Avni Oflaz 阿夫尼



MAJOR

Computer Engineering

ABOUT ME

I am a third grade undergraduate student in an exchange program at Beihang University. I am interested in various types of software and hardware engineering topics such as object oriented programming , robotics and AI accelerators.

STRENGTHS FOR THE PROJECT

- C,C++ languages
- ROS2 and Gazebo simulations
- Applications of Python

RATIONALE

Why this project

A proven teaching case

The same case study used at UPM to teach object-oriented analysis and design — well tested and documented across a full course.

Focus on the process

Requirements, actors and business rules are already defined, so effort goes into learning the engineering process, not inventing a domain.

Full lifecycle coverage

Touches every phase of the course: requirements, use cases, domain modeling, contracts, design diagrams and a design class diagram.

Realistic complexity

A primary actor, a customer, and four external systems (payment authorization, taxes, accounting, HR) — complex enough to be interesting.

THE SYSTEM

What NewEra POS does

NewEra POS is the point-of-sale application a cashier uses to record a customer's purchase, calculate totals and taxes, take payment, update inventory and print a receipt — and, more broadly, to process returns and equipment rentals.

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Use cases: Process Sale, Handle Returns, Process Rental, Start Up

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External systems: payment authorization, taxes, accounting, HR

3

Payment methods: cash, credit card

SYSTEM CONTEXT

Actors & stakeholders

Primary actors

Cashier
Operates the POS terminal for every use case

Customer
Purchases, returns or rents items and pays for them

External systems

Payment Service

Tax Calculator

Accounting System

HR System

NUEVAERA POS

Thank you

Questions & discussion